

JAMES CENAWOOD

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EDUCATION

Cornell University, College of Engineering

August 2024 – May 2028

Bachelor of Science in Computer Science

GPA: 3.9

Selected Coursework: Probability & Statistical Inference (A+), Stochastic Processes (A+), Linear Algebra, Discrete Structures; Machine Learning and Analysis of Algorithms in progress

EXPERIENCE

Northwell Health | *Data Science Intern*

June 2026 – August 2026

- Built a LangGraph NL-to-SQL agent with deterministic schema/SQL validation and no patient-data access
- Raised schema recall@20 from 53.8% to 74.9% across ~250 analyst queries by fusing BM25-ranked SQLite FTS and dense-vector retrieval over 40,551 Epic schema files and 482,130 indexed chunks
- Built the stratified evaluation benchmark that attributed 82.4% of top-5 retrieval misses to vocabulary mismatch rather than ranking depth across 74 answerable questions
- Tied for first in Northwell's intern Kaggle competition with a 0.81014 public-leaderboard accuracy score using an Optuna-tuned CatBoost and TabNet ensemble

Cornell Physical Intelligence | *Software Subteam Lead*

December 2025 – Present

- Owned the research agenda and core implementation for an eight-engineer software subteam developing mapless visual navigation and collective-thrust/body-rate control for Anduril's AI Grand Prix; passed Virtual Qualifier 1 with a time of 17.2s
- Prevented invalid policies from reading unavailable race-time state and automated simulator launch, scoring, telemetry and visual capture, and summaries with a Python/PowerShell experimentation harness for coding agents

Cornell Extended Reality | *Software Team Member*

August 2025 – Present

- Designed a persistent, person-specific memory for an AI smart-glasses assistant through a typed, multi-tenant FastAPI/PostgreSQL backend and natural-language identity resolver over episodes, time-scoped facts, summaries, and relationship graphs

SELECTED RESEARCH & PROJECTS

Surpassing the Leader: Exact Stochastic-Game Solver

github.com/palerdr/SURPASSING-THE-LEADER

- Formalized Usogui's final two-person gamble as a finite simultaneous-move zero-sum stochastic game; authored the accompanying paper and implemented the game and verification pipeline across C++, Rust, Python, and OCaml
- Proved a value-preserving quotient reducing 5,267,489,760 reachable states to 289,374,121 equivalence classes ($18.203\times$) and a strictly increasing potential ordering them into 1,201 layers, eliminating graph search and reachability analysis
- Completed the full backward-induction tablebase in 49 seconds on a 15-core laptop from a single 480-line Python file, versus a projected 3.7–5.1-year single-core search; bounded maximum unilateral policy improvement below 6.005×10^{-4} and derived a 54.493% first-Dropper win probability

AlgoSplit: Quantitative Hypertrophy Modeling

algo-split.vercel.app

- Formalized a weekly hypertrophy stimulus-fatigue model from hypertrophy research for low-level insights into training
- Designed an auditable 29-region stimulus research engine for comparing and analyzing training splits before they are run, defining assumptions for movement-pattern-to-muscle attribution, resistance profiles, volume, and scheduling
- Increased model-scored net weekly stimulus by 35% on average across 100+ user-submitted training splits through generated exercise, volume, and scheduling optimizations
- Operationalized free-text lift names as a 34-class feature representation mapping exercises to movement patterns, target muscles, resistance profiles, and fatigue features; achieved 100% exact-match accuracy and 1.00 macro-F1 across 196 curated strings, then shipped the model through authenticated web and iOS clients

Sentinel: Intelligent Waiting-Room Triage | **1st Place, URM x WICC x Gemini Hackathon**

sentinel.thekoppe.com

- Won first place by building a QR-based waiting-room triage platform whose deterministic engine converts conversational intake, symptoms, vitals, and wait time into ESI estimates, risk scores, follow-up actions, and a live queue
- Designed emergency overrides, confidence/noise penalties, promotion hysteresis, and delayed demotion; constrained the LLM reviewer to within-band reordering, escalation suggestions, and targeted follow-up questions

TECHNICAL SKILLS

Languages: Python, C++, Rust, SQL, TypeScript, OCaml

Libraries/Frameworks: PyTorch, scikit-learn, NumPy, pandas, Polars, FastAPI, LangGraph, OpenCV, React, Expo

Infrastructure/Tools: PostgreSQL, SQLite, BigQuery, Docker, Supabase, Vercel, Optuna, Weights & Biases, Git, LaTeX

Interests: Poker (Cornell Poker Club: 3rd, Citadel \$2K Guaranteed; 3rd, Jane Street Freeroll), Game Design, 3D Animation, VR